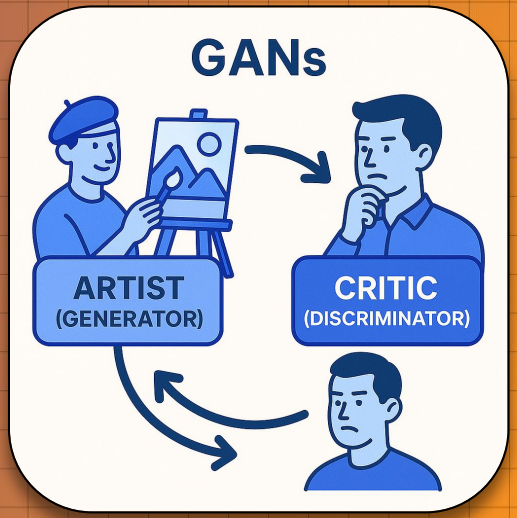
How AI Generates Images

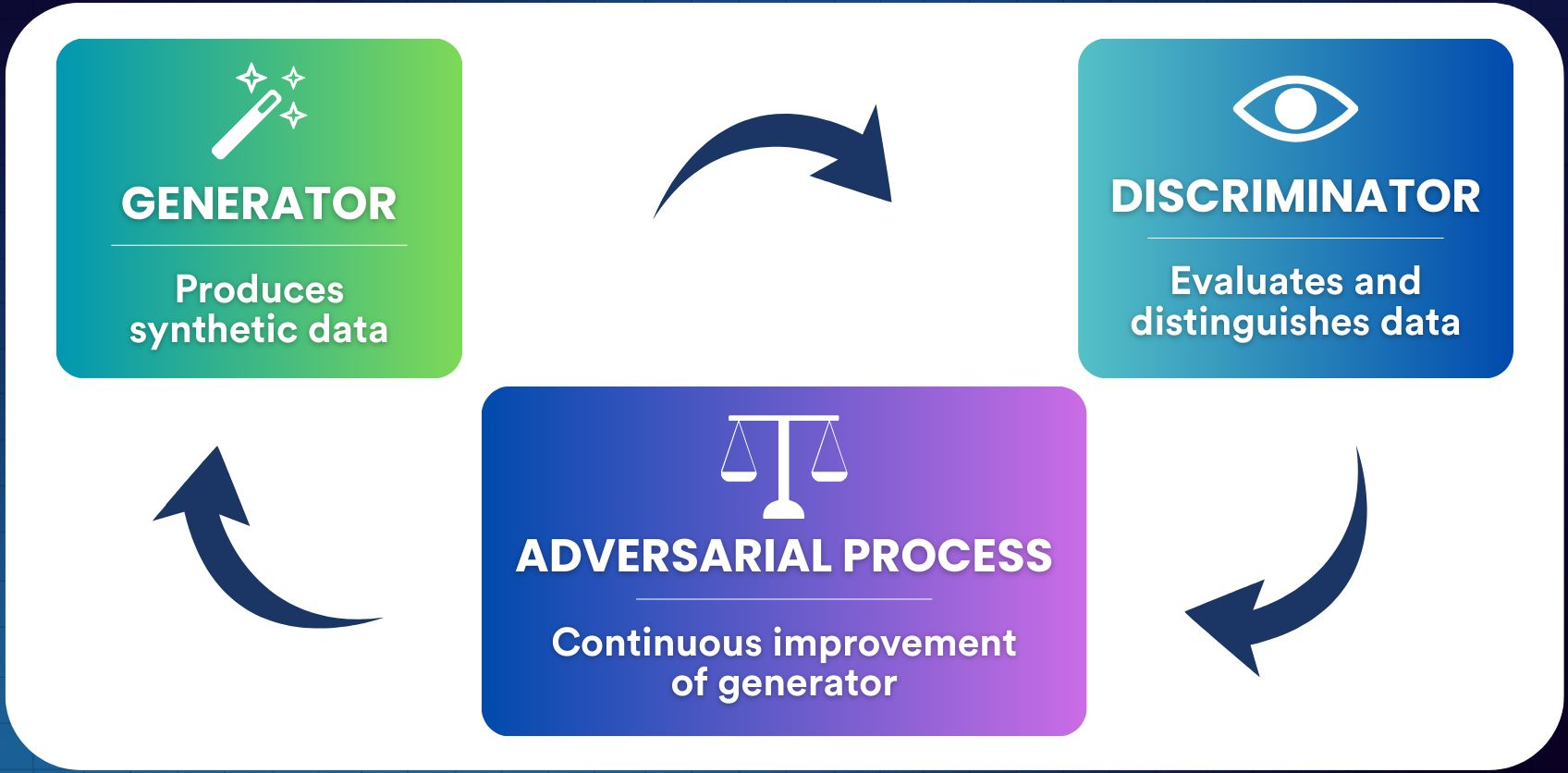
- Trained on millions of images: AI learns patterns, colors, shapes
- Builds from scratch, pixel by pixel—no templates, no cut-and-paste
- Starts with static noise, gradually forms a complete, vivid picture

GANs: The Creative Duel

- Generative Adversarial Networks (GANs)
- Generator: produces fake images designed to look real
- Discriminator: tests each image, spotting fakes vs. real
- A continuous feedback loop that sharpens both—like an endless art battle

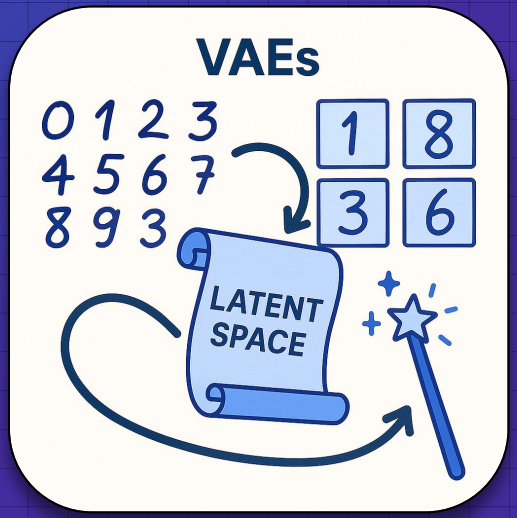


GENERATIVE ADVESARIAL NETWORKS (GANs) STRUCTURE

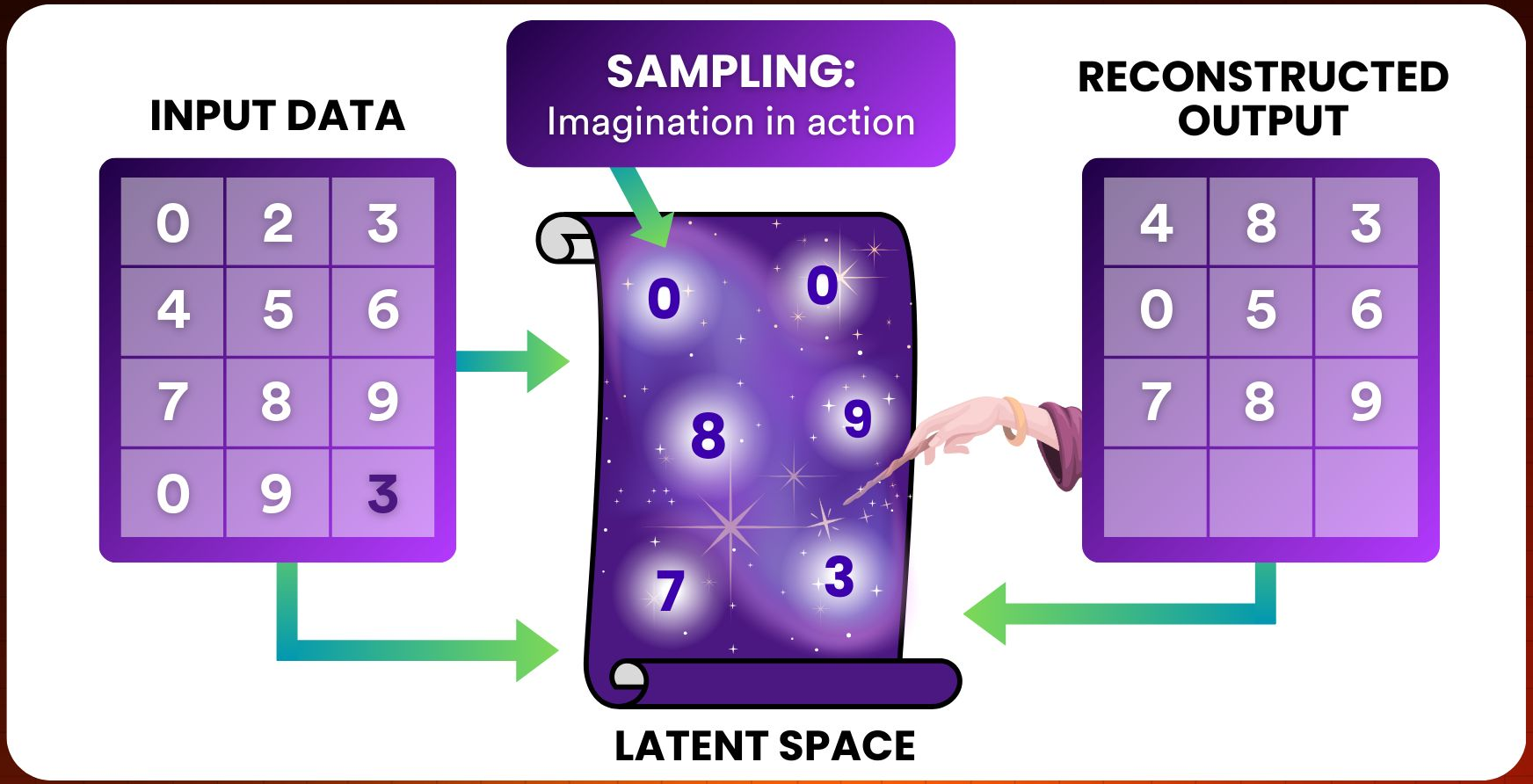


VAEs: Compress, Imagine, Rebuild

- Encodes data into latent space—an abstract, compressed version
- Samples points from that space to create new, realistic images
- Great for generating smooth, controlled variations—never just copies

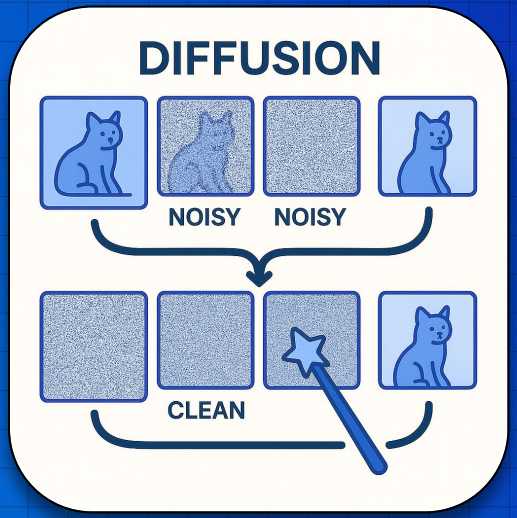


VARIATIONAL AUTOENCODERS (VAEs)

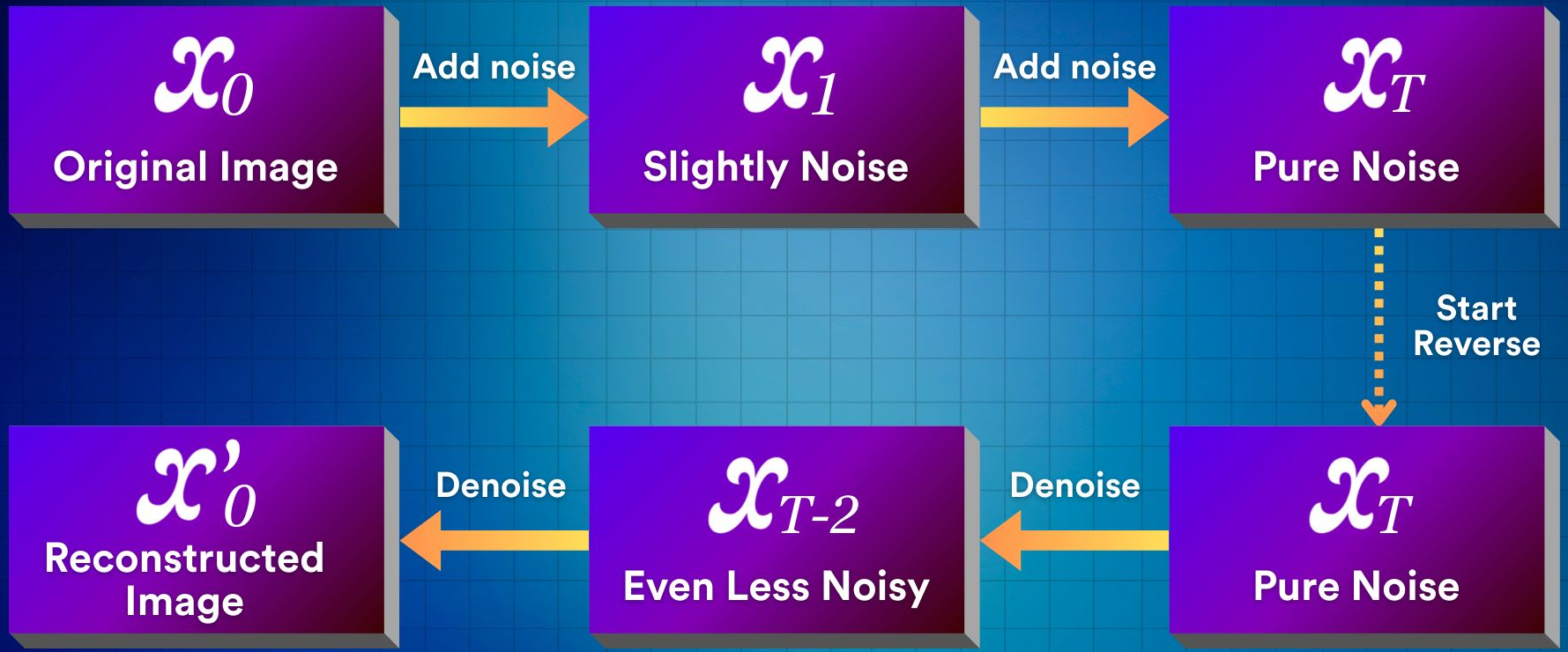


Diffusion Models: From Noise to Art

- AI learns both to destroy (add noise) and rebuild (remove noise)
- Begins with pure static and sculpts it into clear, detailed images
- Excels at producing high-quality, diverse outputs—but slower than GANs



DIFFUSION PROCESS



How Text Becomes an Image

- AI decodes your words: turns prompts into color, shape, and mood
- Clearer, more detailed prompts = sharper and more accurate images
- Think of it like a chef: your prompt is the recipe for the final dish

a minimalist watercolor painting of a sunrise over

Models in Action: Visual Metaphors

- GANs: speed painters—quick, convincing results, but can repeat
- VAEs: architects—compress & rebuild with creative flexibility
- Diffusion: sculptors—carve detailed images from pure noise

Today’s Top AI Image Tools

- **MidJourney:** stylized, dreamlike visuals, web based app
- **4o Image Generation:** precise, flexible prompt understanding; built into ChatGPT
- **Stable Diffusion:** fully customizable, open-source, community-powered

FEATURE	MIDJOURNEY	CHATGPT-4o (Image Gen)	STABLE DIFFUSION
Deployment	Web-based interface	Integrated into ChatGPT interface	Local or cloud (open-source)
Model Type	Proprietary diffusion-based model	Proprietary multimodal model with image generation	Open-source diffusion-based model
Strengths	Artistic, stylized visuals with cinematic flair	Excellent prompt understanding, fast, in-chat results	Fully customizable, privacy-friendly
Weakness	Limited fine-tuning, aesthetic over realism	Pay-per-use tokens, less control over style	Setup complexity, higher hardware requirements
Best Use Cases	Concept art, moodboards, fantasy aesthetics	Conversational image creation, fast prototyping	Domain-specific projects, advanced control
Pricing	Subscription-based, with limited free access	Pay-per-use (via ChatGPT Pro or API tokens)	Free to use, local hardware or cloud costs